Cluster Analysis

Clustering background

Introdfuction

An unsupervised learning technique

Correct number of clusters is unknown

Distance / Similarity measurement (算距離)

Names	Formula	
Euclidean distance	$\lVert a-b \rVert_2 = \sqrt{\sum_i (a_i - b_i)^2}$	兩點之間直線距離
Squared Euclidean distance	$\lVert a-b \rVert_2^2 = \sum_i (a_i - b_i)^2$	兩點之間直線距離不要開根
Manhattan distance	$\lVert a-b \rVert_1 = \sum_i \lvert a_i - b_i \rvert$	兩點之間X軸距離+Y軸距離
Maximum distance	$\lVert a-b \rVert_\infty = \max_i \lvert a_i - b_i \rvert$	
Mahalanobis distance	$\sqrt{(a-b)^\top S^{-1}(a-b)}$ where S is the Covariance matrix	

Clustering Algorithm Objective

Minimize Intra-clusters

Maximize Inter-clusters

Hierarchical algorithms

Introduction

Agglomerative 聚合法: 從每個資料一群逐漸結合成多個資料一群

Divisive 分割法：全部資料一群分割成多個資料一群

最佳化 Tightness

Silhouette Index Algorithm

a(i): is the average distance in a clustered group $a(i) = \frac{1}{|C_i| - 1} \sum_{j \in C_i, i \neq j} d(i,j)$

b(i): minimum average distance between i and other group $b(i) = \min_{i \neq j} \frac{1}{|C_j|} \sum_{j \in C_j} d(i,j)$

$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$ if |C| > 1, otherwise $s(i) = 0$

GS_u : Total score (higher is better) $GS_u = \frac{1}{c} \sum_{j=1}^{c} S_j$

K-Means

Partitioning Algorithms that split data into K clusters

Cluster based on centroid (重心) $\bar{\mu}(c) = \frac{1}{c} \sum_{x \in c} x$

Steps: 1. Select ramdom K seeds as different start

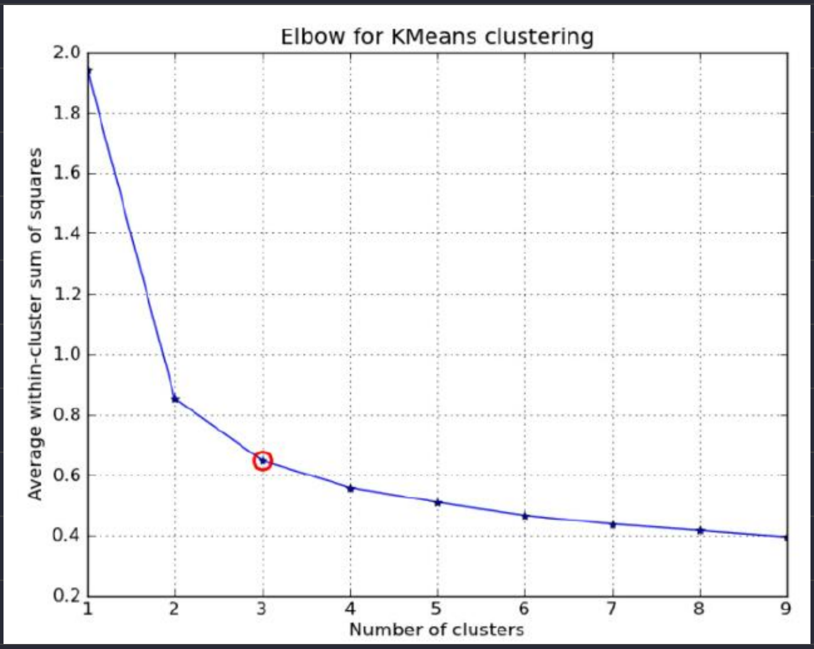
2. Assign every data to the cluster based on $\bar{\mu}$

3. Calculate new centroid of current K clusters

4. Repeat step 1 to step 4 until centroid / partition not changed

Seed choice can significantly affect the result, try with different random seed

How to choose K:

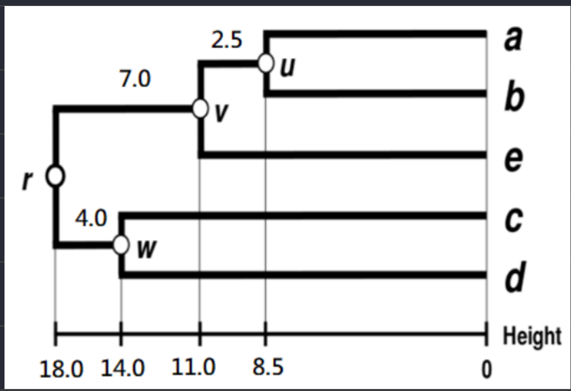


Hierarchical Clustering

Can be execute with bottom-up and top-down

Don't need to specify K

Dendogram: Hierarchical Clustering



Cluster distance measure

Single-link：Distance of the “closest” points

Complete-link：Distance of the “farthest” points

Centroid：Distance of the centroids (centers of gravity)

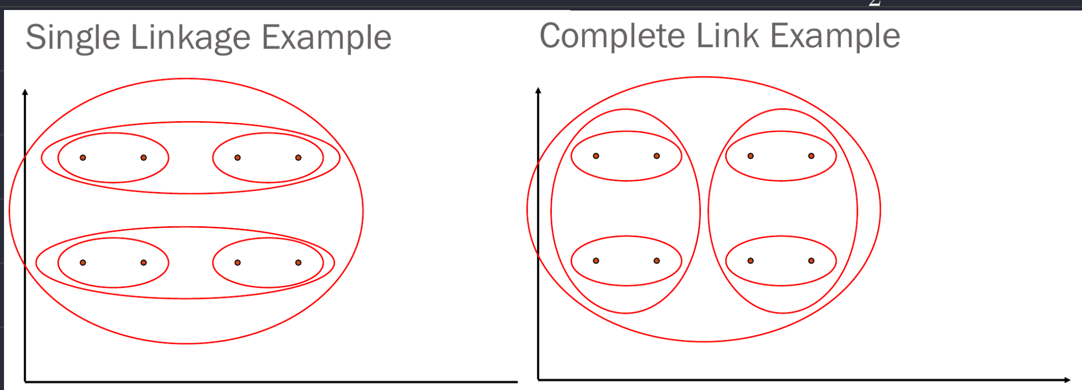
Average-link：Average distance between pairs of elements

UPGMA (unweighted pair group method with arithmetic mean)：

Cluster 中的數據數量不影響其中點位的權重 $d(A,B)=\frac{1}{|A|\cdot|B|}\sum_{a\in A}\sum_{b\in B}d(a,b)$

WPGMA (Weighted Pair Group Method with Arithmetic Mean)

Cluster 中的數據數量案比例影響其中點位權重 $d(i\cup j,k)=\frac{1}{2}(d(i,k)+d(j,k))$



DBSCAN（Density-Based Spatial Clustering of Application with Noise）

Introduction

基於密度與距離測量的演算法

r：半徑　　MinPoints：最少資料點

Type of points

Core Points (核心點)：在 r 範圍內包含自己有 MinPoints 以上的資料點

Border Points (邊界點)：在 r 範圍內只有 < MinPoints 的資料點 ，但是包含核心點

Noise (雜訊)：不屬於以上兩者

Density Reachable (密度可達)

直接密度可達（directly density-reachable）

點 A 為核心點, 點B任意，且 $|A-B| < r$，則 $A \longrightarrow B$ 為直接密度可達

密度可達（density-reachable）：

點 A 為核心點, 點B任意，雖 $|A-B| \geq r$ 但是可以通過其他點間接連線，則 $A \longrightarrow B$ 為密度可達

Clustering Methods

Hard clustering：分群不會重疊，每筆資料只有一種類別

Soft clustering：每筆資料會有多種類別與信心程度

Expectation Maximization (EM) 期望最大化演算法

一種 Soft Clustering 的方法

1. 假設隨機起始位置
2. 進入 E-Step 計算每個資料點在個群組中的機率
3. 進入 M-Step 調整參數讓 $Q(\theta, \theta')$ 增加
4. 反覆 2 到 3 步驟並找到能讓 Q (評分函數) 最大化的 θ